

# From stochastic gradient descent to Adam

1. SGD:  $\theta^{(t+1)} \leftarrow \theta^{(t)} - \alpha \nabla L(\theta^{(t)})$   
→ gradient over a batch

2. Momentum:

1.  $m^{(t+1)} \leftarrow \alpha m^{(t)} - \eta \nabla L(\theta^{(t)})$

2.  $\theta^{(t+1)} \leftarrow \theta^{(t)} + m^{(t+1)}$

pros: 1. accelerating

example 1:

if  $\nabla L(\theta^{(t)}) = g$  is a constant, then

$$m^{(1)} = -\eta g$$

$$m^{(2)} = -\eta g - \eta^2 g$$

$$m^{(3)} = -\eta g - \eta^2 g - \eta^3 g$$

⋮

$$m^{(n)} = -(\eta + \dots + \eta^n) g$$

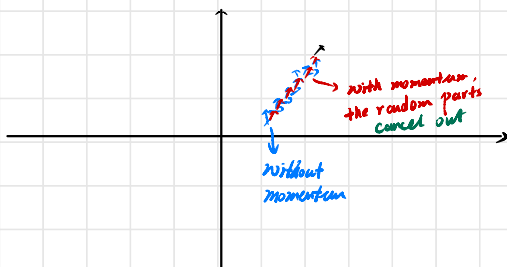
$$= -\frac{\eta(1-\eta^n)}{1-\eta} g$$

e.g. if  $\eta = 0.9$ , then

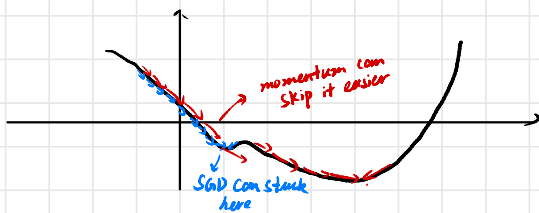
$$m^{(n)} \rightarrow -g g \text{ when } n \rightarrow \infty$$

→ accelerate 10 times (plus the gradient)

example 2:



2. skip local minimum



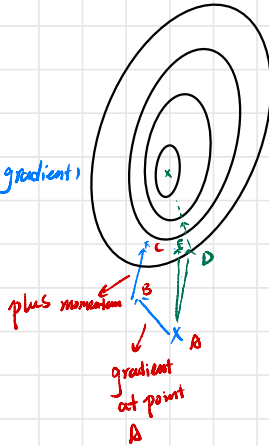
### 3. Nesterov Accelerated Gradient

$$1. m^{(t+1)} \leftarrow \alpha m^{(t)} - \eta \nabla L(\theta^{(t)} + \alpha m^{(t)})$$

$$2. \theta^{(t+1)} \leftarrow \theta^{(t)} + m^{(t+1)}$$

→ observation here:  
in general the momentum  
vector will point to the  
right direction (i.e. minimum)

momentum: from A to B  
(by taking gradient)  
and then B to C  
(by adding momentum)



Nesterov: apply momentum first, from  
A to D, take gradient at point D  
(which in general points to the  
minimum better) and then  
add the gradient and momentum

### 4. AdaGrad

$$1. S^{(t+1)} \leftarrow \beta S^{(t)} + (1-\beta) \nabla L(\theta^{(t)}) \odot \nabla L(\theta^{(t)})$$

$$2. \theta^{(t+1)} \leftarrow \theta^{(t)} - \alpha \nabla L(\theta^{(t)}) \oslash \sqrt{S^{(t+1)} + \epsilon}$$

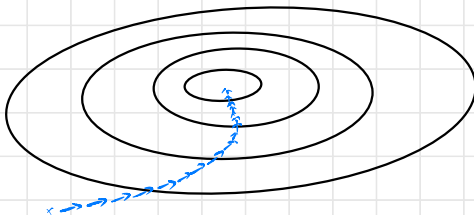
→ pointwise prod

→ pointwise division

Observation here: when the gradients in one direction is much steeper in another.

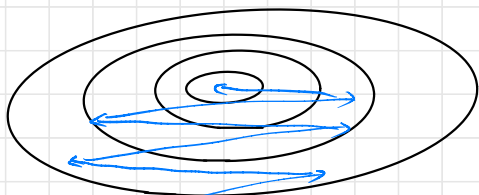
it is very hard to choose one learning rate

- (1) make good progress in both direction
- (2) stable



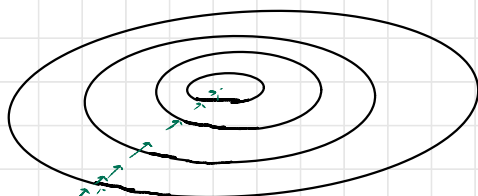
gradient is steep  
horizontally

→ choose good learning rate for horizontal direction (SGD)  
would be very slow  
because vertical direction  
has much smaller gradient



gradient is steep horizontally

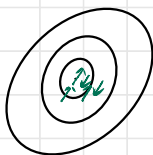
→ choose good learning rate for vertical direction (SGD) would be very unstable



gradient is steep horizontally

→ AdaGrad: scale both direction. So it move the same distance along each direction

Cons: It does not converge to the exact minimum but instead oscillates around it (because the moving distance is the same, i.e. it only cares about directions, agnostic about the exact value of the gradient, so it doesn't know the gradient is small and need to stop)



5. Adam: Combines momentum and AdaGrad

$$1. m^{(t+1)} \leftarrow \beta_1 m^{(t)} + (1 - \beta_1) \nabla L(\theta^{(t)})$$

momentum   
 it helps AdaGrad converge

$$2. S^{(t+1)} \leftarrow \beta_2 S^{(t)} + (1 - \beta_2) \nabla L(\theta^{(t)}) \otimes \nabla L(\theta^{(t)})$$

adaptive learning rate

$$3. \hat{m}^{(t+1)} \leftarrow \frac{m^{(t+1)}}{1 - \beta_1^t}$$

$$4. \hat{S}^{(t+1)} \leftarrow \frac{S^{(t+1)}}{1 - \beta_2^t}$$

} bias correction: because  $m$  and  $S$  initialized at 0, they are biased towards 0

$$5. \theta^{(t+1)} \leftarrow \theta^{(t)} + \alpha \hat{m}^{(t+1)} \oslash \sqrt{\hat{S}^{(t+1)} + \epsilon}$$